

Federated Generative Modeling for 3D Furniture Layout: Impact of Non-IID Data Heterogeneity

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Abstract

In the home furnishing and interior design industry, multiple companies increasingly require data-driven layout models to improve design efficiency and personalization, yet raw design data are rarely shared across institutions because they often encode commercially sensitive assets such as designer know-how, customer preferences, brand positioning, and proprietary layout strategies. This paper investigates the convergence behavior and effectiveness of federated generative models for automated 3D furniture placement under Non-IID data distributions. Furniture layouts are formulated as structured sets of objects encompassing semantic and geometric attributes. To simulate real-world collaborative design scenarios, two distinct forms of data heterogeneity are modeled: room-type-based label shift and style-based feature shift. Under a unified federated protocol with controlled experimental settings, we conduct a systematic analysis from three perspectives: convergence dynamics and training stability, synthesis quality and layout feasibility, and collaborative effectiveness under practical communication constraints. We further examine the effects of system parameters, including communication frequency and the number of local update steps, on training behavior. The results provide empirical evidence and practical implications for deploying federated generative models for collaborative indoor design under privacy-preserving constraints.

Keywords: 3D Indoor Layout Generation, Generative Modeling, Federated Learning, Non-IID Data Heterogeneity

1. Introduction

Automated 3D furniture layout generation has substantial practical value for improving interior design iteration, reducing manual design costs, and supporting personalized space design at scale. Recent generative methods have shown the ability to synthesize reasonable and credible furniture layouts from room contours or room-type distributions. In practical deployment, however, interior design data are usually held separately by design firms, home furnishing platforms, and furniture manufacturers. Due to commercial confidentiality, user privacy, and regulatory requirements, these parties often cannot aggregate raw samples centrally for unified model training. This creates an unavoidable data silo problem, limiting data coverage and generalization for layout synthesis models that need to integrate cross-institutional data and evolve over time, thereby compromising the reliability and cross-brand usability of automated layout systems.

Federated learning offers a viable paradigm for cross-institutional collaborative model training: each participant retains its local data and trains on site, while only model updates are exchanged for aggregation. For structured 3D layout synthesis, this setting is particularly challenging because the task depends not only on object distributions, but also on spatial priors, geometric constraints, and combinatorial relations among furniture instances. In realistic scenarios, local datasets tend to exhibit systematic Non-IID characteristics due to differences in design preferences and regional contexts. For example, some companies may mainly accumulate bedroom layouts, whereas others may focus on living-room or dining-room projects; even within the same room type, local design teams may differ in compactness, spacing, density, and object arrangement patterns.

This paper focuses on federated generative modeling for 3D furniture layout synthesis under given room constraints. The goal is to generate layouts that remain distributionally consistent with real designs while satisfying boundary, physical, and functional constraints. To investigate this problem under heterogeneous multi-source client data, we organize the study around the following research questions:

- **RQ1:** In automated 3D furniture layout generation, how do room type-induced label shift and style-induced feature shift affect the training behavior and convergence characteristics of federated generative models, particularly when such heterogeneity arises from differences in

industrial specialization, regional demand, and cultural design preference?

- **RQ2:** Under federated training for automated furniture placement, how do style variation and room-type data heterogeneity affect the quality, plausibility, and diversity distribution of synthesized layouts?
- **RQ3:** To what extent can federated training effectively integrate data contributions from diverse clients to improve automated furniture placement capability under practical constraints such as limited communication and local computation?

The remainder of this paper is organized as follows. Section 2 introduces the background and related work. Section 3 presents the data heterogeneity modeling and federated training setup. Section 4 reports the experiments in correspondence to the proposed research questions. Section 5 discusses the findings, limitations, and practical implications by revisiting RQ1–RQ3. Section 6 concludes the paper.

2. Related Work

Cross-institutional collaborative modeling is highly demanded across industrial settings such as interior design [1]. However, datasets accumulated by different organizations are difficult to consolidate into a single corpus due to commercial confidentiality and privacy constraints, rendering the traditional paradigm of unified training increasingly untenable [2, 3]. In this context, Federated Learning (FL) offers a viable solution to multi-institutional data silos [4, 5]. The standard FedAvg algorithm integrates models through iterative local training and global aggregation [4]. While effective on homogeneous data, its performance is highly sensitive to statistical heterogeneity. Prior work has shown that pronounced Non-IID shifts make federated optimization susceptible to convergence oscillation and degradation [6, 7]. Practical deployments further reveal that such heterogeneity is often intertwined with communication and device constraints [2, 8, 9].

At the application level, generative modeling for 3D indoor scene synthesis has evolved from sequential arrangement to diffusion-based paradigms. ATISS adopts a Transformer architecture for serialized furniture placement [10],

while DiffuScene further improves synthesis realism through progressive denoising [11]. Recent denoising and score-matching methods, such as DeBaRA [12] and SemLayoutDiff [13], introduce stronger geometric and semantic controllability. These methods provide robust tools for modeling structured layout distributions but are predominantly evaluated in centralized training settings.

To bridge the gap between decentralized data and generative performance, federated generative learning has been explored [14, 15]. Researchers have utilized distributional discrepancy measures, such as Earth Movers Distance and KNN-based metrics, to characterize data drift [7, 16]. Personalized FL approaches like PFL-GAN [17] and FedPer [18] further suggest using shared-private decomposition to handle client heterogeneity. Despite these advances, existing efforts predominantly focus on final visual quality in medical or natural imaging [19, 20]. There remains a lack of systematic analysis regarding how structured 3D layouts behave under specific Non-IID partitions, especially with respect to the relative impact of label-space versus feature-space heterogeneity on federated generative training.

3. Methodology

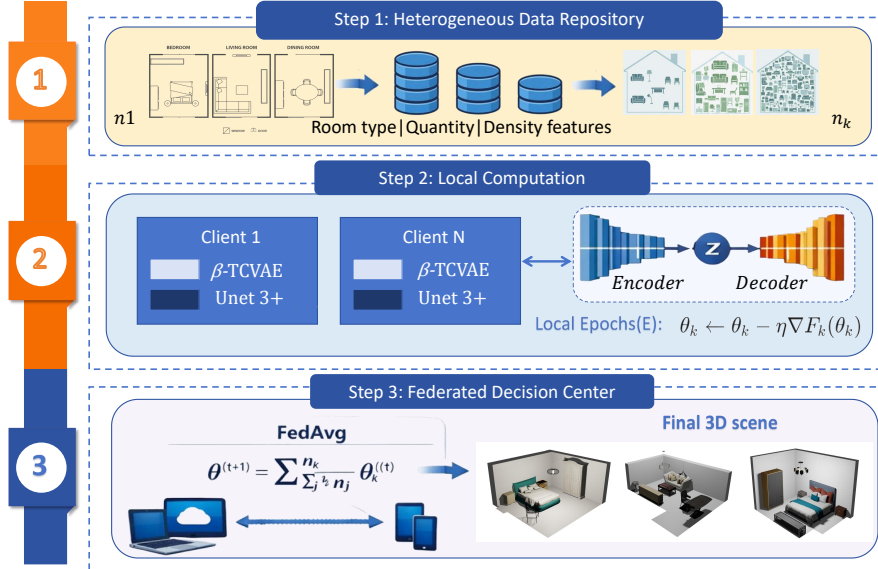


Figure 1: Overview of the federated generative layout modeling framework.

To model structured 3D furniture layouts under federated data constraints, this study represents indoor scenes as structured object sets and implements layout generation through an Enhanced β -TCVAE generator, category-wise alignment, a UNet3+-based relational branch, and FedAvg-based aggregation.

3.1. Federated Generative Layout Modeling under Practical Constraints

An indoor scene is expressed as a structured set

$$S = \{o_i\}_{i=1}^N, \quad (1)$$

where N denotes the number of furniture instances in a room and may vary across scenes. Each instance is parameterized as

$$o_i = (c_i, t_i, s_i, r_i), \quad (2)$$

where $c_i \in \{1, \dots, K\}$ denotes the semantic category, $t_i \in \mathbb{R}^3$ the 3D translation, $s_i \in \mathbb{R}^3$ the object size, and $r_i \in \mathbb{R}$ the azimuthal orientation.

This formulation encodes the room-level structure of 3D-FRONT as a variable-cardinality collection of semantic and geometric descriptors, providing the structured input space for subsequent generative modeling.

3.1.1. Enhanced β -TCVAE generator

After each 3D-FRONT scene is converted into the above object-set representation, the semantic and geometric attributes of furniture instances are organized as the generator input x . Based on this processed layout representation, the generator adopts an Enhanced β -TCVAE architecture for latent representation learning. Unlike a plain VAE parameterization, the present implementation combines hierarchical sparse transformations, residual refinement, and latent regularization decomposition to better model structured object sets. Specifically, the encoder maps x into latent parameters $(\mu, \log \sigma^2)$, from which latent codes z are sampled via the reparameterization trick. The decoder then reconstructs object-wise layout attributes through structured upsampling and sparse linear projection.

Given input x , the generator objective is defined as

$$\mathcal{L}_{gen} = \mathcal{L}_{rec} + \alpha \mathcal{L}_{MI} + \beta \mathcal{L}_{TC} + \gamma \mathcal{L}_{KLD}, \quad (3)$$

where $\mathcal{L}_{rec}$ denotes reconstruction loss, $\mathcal{L}_{MI}$ is the mutual-information term, $\mathcal{L}_{TC}$ is the total-correlation term, and $\mathcal{L}_{KLD}$ is the dimension-wise divergence

term. In our implementation, $\alpha = 1.0$, $\beta = 6.0$, and $\gamma = 1.0$. To improve stability during optimization, KL-related regularization is annealed in the early stage of training.

The decomposition of latent regularization encourages a more factorized representation, which is beneficial when different clients exhibit heterogeneous room-type composition or stylistic spatial patterns. Architecturally, the encoder-decoder pipeline incorporates sparse linear blocks, residual blocks, and channel-sensitive transformations to balance latent abstraction with layout-specific structural fidelity.

3.1.2. Category-wise alignment and reconstruction

A key difficulty in set-structured layout generation is that reconstructed object slots do not have a fixed one-to-one ordering with the ground-truth furniture instances. To address this issue, reconstruction is computed after category-wise linear assignment. Within each furniture category, a distance matrix is constructed between reconstructed outputs and ground-truth instances, and optimal matching is obtained through linear assignment.

After alignment, the reconstruction objective is decomposed into category-related and angle-related terms. For object attributes excluding angle classes, Smooth L1 loss is used:

$$\mathcal{L}_{attr} = \text{SmoothL1}(\hat{x}_{attr}, x_{attr}). \quad (4)$$

For rotation modeling, the angle is represented through a discrete class together with a residual term. Accordingly, the angle-related loss is defined as

$$\mathcal{L}_{angle} = \mathcal{L}_{cls}^{angle} + \lambda_{res} \mathcal{L}_{res}^{angle}, \quad (5)$$

where $\mathcal{L}_{cls}^{angle}$ is a cross-entropy term for angle classification and $\mathcal{L}_{res}^{angle}$ is a regression term for angle residual prediction. The overall reconstruction component is then written as

$$\mathcal{L}_{rec} = \mathcal{L}_{attr} + \mathcal{L}_{angle}. \quad (6)$$

This matching-based formulation is important because layout synthesis involves multiple furniture instances of the same category, and direct slot-wise supervision would otherwise introduce mismatch noise into training.

3.1.3. UNet3+-based relational constraint network

To complement object-wise synthesis, we introduce a UNet3+-based relational branch for spatial regularization. This branch does not function merely as a binary discriminator in the conventional GAN sense. Instead, it operates as a multi-scale relational prediction network that explicitly models pairwise spatial structure among furniture objects.

For each scene, object features are first converted into pairwise combinations. The relational input includes parsed object attributes together with category-related indicators, allowing the network to reason jointly over pairwise geometry and object-type dependency. The recombined tensor is then processed by a UNet3+ architecture with multi-scale feature fusion. The network outputs relation predictions at multiple decoder stages, which are jointly supervised during training.

The spatial objective is formulated as

$$\mathcal{L}_{spatial} = \lambda_1 \mathcal{L}_{ind} + \lambda_2 \mathcal{L}_{trans_cls} + \lambda_3 \mathcal{L}_{trans_res} + \lambda_4 \mathcal{L}_{rot} + \lambda_5 \mathcal{L}_{size} + \lambda_6 \mathcal{L}_z, \quad (7)$$

where $\mathcal{L}_{ind}$ supervises relation indicators, $\mathcal{L}_{trans_cls}$ and $\mathcal{L}_{trans_res}$ model discretized relative translation and its residuals, $\mathcal{L}_{rot}$ models relative orientation, $\mathcal{L}_{size}$ supervises size similarity, and $\mathcal{L}_z$ constrains vertical offset. Through these terms, the relational branch regularizes the synthesized layouts beyond marginal object prediction and encourages coherent object arrangement in global space.

3.1.4. Federated training formulation

Let $\mathcal{D}_k$ denote the local dataset of client k , where the data distribution may differ across clients due to room-type composition, style preference, or sample quantity imbalance. At each communication round, client k updates its local model parameters by minimizing

$$\mathcal{L}^{(k)} = \mathcal{L}_{gen}^{(k)} + \eta \mathcal{L}_{spatial}^{(k)}, \quad (8)$$

where η controls the contribution of the relational branch. After local training, the server aggregates model parameters using weighted averaging:

$$\theta^{(t+1)} = \sum_{k=1}^K \frac{n_k}{\sum_{j=1}^K n_j} \theta_k^{(t+1)}, \quad (9)$$

where $\theta_k^{(t+1)}$ denotes the updated parameters from client k after round t , and n_k is the number of training samples on that client.

It should be emphasized that the present study does not propose a new federated optimization algorithm. Instead, FedAvg is used as a controlled baseline coordination mechanism, so that the observed differences in convergence and synthesis quality can be attributed primarily to heterogeneity in client data rather than to changes in the aggregation rule.

3.2. Realistic Data Heterogeneity Modeling among clients for Federated Generative Scene Layouts

3.2.1. Mathematical Definition of Data Heterogeneity

In federated learning, the Non-IID property refers to inconsistencies in data distributions across clients, which can be expressed as:

$$p^{(k)}(y = i) \neq p(y = i). \quad (10)$$

Here, $p^{(k)}(y = i)$ denotes the local data distribution of the k -th client, while $p(y = i)$ denotes the global data distribution after aggregation [16]. In this work, the original 3D-FRONT dataset is used to emulate authentic furniture layout design data, and data heterogeneity is modeled and analyzed from two perspectives: category distributions and continuous geometric feature distributions.

3.2.2. Room-based Label Shift

In real federated layout generation tasks, room type is a pivotal determinant of the composition and quantity distribution of furniture categories. Different room functions correspond to distinct furniture distributions. For example, bedroom scenes are primarily associated with beds and wardrobes, whereas living room scenes are mainly associated with sofas and TV cabinets. In industrial scenarios, different business lines may serve specific room types over long periods of time, and such specialization can also be shaped by regional housing demand and cultural design preference. As a result, local data sources tend to exhibit clear Non-IID characteristics in terms of spatial functional semantics.

The room-based label shift considered in this paper is defined as:

$$p^{(k)}(c) \neq p(c), \quad k = 1, \dots, K, \quad (11)$$

where $p^{(k)}(c)$ denotes the category distribution of the k -th client, and $p(c)$ denotes the global distribution.

To characterize this type of heterogeneity, we analyze the room-type composition of 3D-FRONT. 3D-FRONT is a representative indoor layout dataset that contains multiple room types, making it well suited for simulating room-based client heterogeneity.

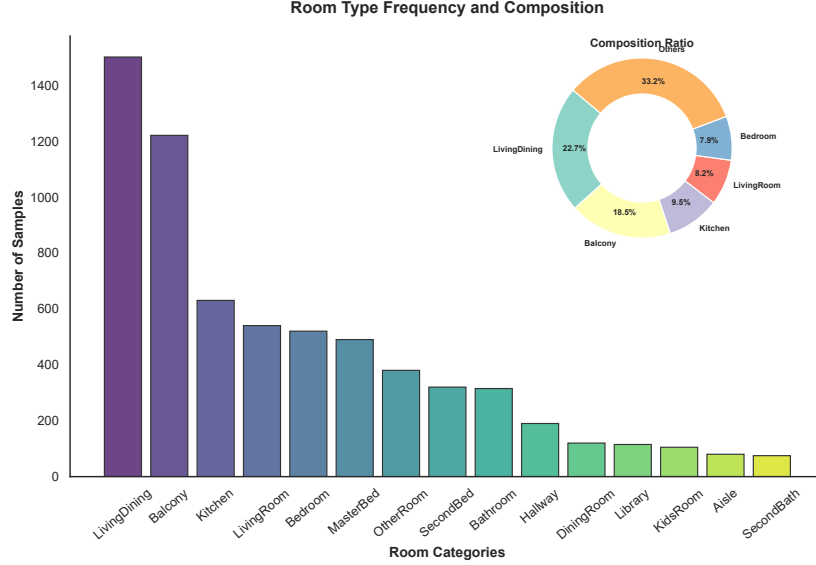


Figure 2: Statistical analysis of room-type labels in the 3D-FRONT dataset.

As shown in Figure 2, the sample sizes of different room types exhibit a pronounced long-tail distribution. This observation indicates that room-function imbalance is an inherent characteristic of realistic indoor layout data rather than merely an experimental artifact. Based on this property, we treat room-type subsets in 3D-FRONT as proxies for heterogeneous business data sources and construct a room-type-based partitioning strategy to reproduce the Non-IID challenge caused by differences in business scope in federated learning.

3.2.3. Style-based Feature Shift

Beyond category distributions, interior design style primarily affects the distribution of furniture in continuous attribute space, including size proportions, spatial density, relative distances, and orientations. Different room styles exhibit systematic differences in these features. Under a style-based partitioning strategy, clients are grouped according to scene style, thereby

inducing feature-skew Non-IID characteristics in continuous-variable distributions.

In this paper, the layout compactness indicator is selected as the continuous-feature shift metric to characterize the influence of style on space utilization strategies:

$$\text{density} = \frac{\sum \text{footprint}(obj)}{\text{floor area}}. \quad (12)$$

The shift in probability mass over the continuous feature space is defined as:

$$p^{(k)}(x_{abs}, x_{rel}|c) \neq p(x_{abs}, x_{rel}|c), \quad (13)$$

where x_{rel} denotes relative furniture positions and arrangement density, x_{abs} denotes absolute furniture attributes, c denotes the furniture category condition, and $p^{(k)}$ denotes the distribution of the k -th client.

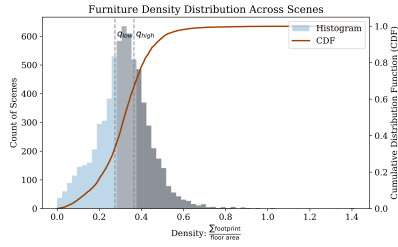


Figure 3: Statistical analysis of style-related density variation in the 3D-FRONT dataset.

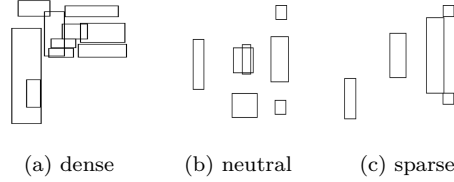


Figure 4: Representative furniture layout patterns under different spatial densities.

As shown in Figure 3, spatial density exhibits a pronounced long-tail distribution, and different layout patterns show substantial deviations in probability mass. To more directly characterize the manifestation of Non-IID characteristics, this paper selects layout deviations caused by different styles within the same room type (living room) to emulate the situation in which samples from different teams in industrial contexts share similar functional semantics but differ in stylistic layout preference.

Based on this observation, we construct a style-based client partitioning strategy to reproduce continuous feature-space shift caused by differences in spatial semantic understanding across clients in federated learning.

3.3. Client Heterogeneity Quantification Measures

To rigorously evaluate the influence of the aforementioned partitioning strategies on federated learning dynamics, we quantify client heterogeneity

across three distinct dimensions: label distribution bias, feature-space (style) bias, and data quantity imbalance.

First, **label skew** is measured using the average label entropy across clients. For a given client k , the entropy $H(Y_k)$ characterizes the diversity of functional room types and is defined as

$$H(Y_k) = - \sum_{i=1}^L p_{i,k} \ln(p_{i,k}), \quad (14)$$

where $p_{i,k}$ denotes the empirical probability of room type i within the local dataset of client k . A lower entropy indicates a higher degree of functional specialization.

Second, we quantify **style skew** by calculating the variance of furniture density distributions, which serves as a proxy for stylistic feature shifts. The style variance, $\text{Var}(\text{style})$, is formulated as the mean squared deviation of local stylistic features from the global average:

$$\text{Var}(\text{style}) = \frac{1}{K} \sum_{k=1}^K (q_{s,k} - \bar{q}_s)^2, \quad (15)$$

where $q_{s,k}$ represents the local style descriptor (e.g., mean furniture density) for client k , and $\bar{q}_s$ denotes the global mean across all K participants.

Finally, **quantity skew** is quantified by the standard deviation of sample counts across the federation. To characterize the extent of volume imbalance, we define the quantity deviation σ_n as

$$\sigma_n = \sqrt{\frac{1}{K} \sum_{k=1}^K (n_k - \mu_n)^2}, \quad \mu_n = \frac{1}{K} \sum_{k=1}^K n_k, \quad (16)$$

where n_k is the number of layout samples held by client k .

4. Experiments

4.1. Analysis of Non-IID Dataset Characteristics and the Rationale for Subset Partitioning

To assess the effectiveness of the proposed model, we conduct empirical evaluation on the large-scale indoor scene synthesis benchmark 3D-FRONT and its associated indoor furniture dataset 3D-FUTURE. To emulate the Non-IID patterns commonly observed in practical federated learning scenarios, we partition the datasets as follows:

- **S0 (baseline)**: A globally unbiased reference where three participants contain equal sample volumes through stratified allocation across all room types and styles.
- **S1 (label skew)**: Simulates functional specialization. Participants are partitioned by room-type bias with naturally occurring imbalances in sample volume. Stylistic heterogeneity remains mild, reflecting design tendencies inherent to specific room functions (e.g., higher furniture density in dining rooms).
- **S2 (feature skew)**: Represents style-oriented expertise. Clients are clustered into three distinct furniture-density groups to induce pronounced stylistic bias, while label distributions (room types) remain aligned with S0.
- **S3 (quantity skew)**: Captures institutional scale disparity. This scheme features highly imbalanced data volumes across participants while maintaining approximately IID label and stylistic distributions.
- **S4 (compound skew)**: A complex bias scenario combining substantial volume imbalance, inconsistent room-type compositions, and pronounced stylistic divergence, reflecting the multifaceted heterogeneity of real-world decentralized data.

Across these configurations, we establish comparative controls spanning from single-dimensional to multi-dimensional data heterogeneity to evaluate the framework’s robustness.

Table 1: Statistics of client heterogeneity under different data partition settings.

Setting	Non-IID type	label_entropy	style_var	n_std
S0	baseline	0.66	0.01	0.0
S1	label skew	0.20	0.01	1664.2
S2	feature skew	0.63	0.22	700.7
S3	quantity skew	0.65	0.01	1950.2
S4	compound skew	0.71	0.05	1551.2

Table 1 shows distinct patterns of data divergence across the five partitioning schemes. S1 (label skew) exhibits the lowest label entropy alongside a substantial sample-volume standard deviation, confirming its role as

a representative of functional label skew. In contrast, S2 (feature skew) is characterized by the highest style variance, implying a more challenging feature-distribution shift compared with the baseline. S3 isolates quantity imbalance while maintaining label entropy and style variance similar to S0 (baseline), whereas S4 (compound skew) reflects compound heterogeneity where both quantity and style distributions deviate from the baseline. These indicators provide a basis for analyzing how different forms of heterogeneity affect stability and synthesis quality.

4.2. Evaluation Metrics

To assess layout quality, we adopt the following evaluation metrics.

4.2.1. Frechet Inception Distance

We measure the distributional discrepancy between the furniture placement maps produced by the model and the real images used during training. In this procedure, the generated placement maps are treated as the target samples, while features extracted from real rendered images in the 3D-FRONT dataset serve as the reference distribution. We use clean-fid as the primary metric to determine whether the produced furniture layouts approach the quality of real data and align with perceptual realism.

4.2.2. Intersection over Union

We compute overlap statistics from the model outputs by measuring the spatial overlap ratio, where a value of 0 indicates no collision. This metric is used to examine whether the produced scenes satisfy physical constraints and retain functional viability in real-world settings.

4.3. Assessing Collaborative Effectiveness through Local Update Control

In federated training, the number of local update steps E significantly modulates training stability and collaboration efficiency. Empirical observations indicate that when $E = 1$, the model attains the lowest training loss and exhibits the smoothest convergence trajectory; as E increases, the loss curve reaches convergence more rapidly, but is subsequently accompanied by oscillation and a higher final loss. To quantify this effect, the same group is trained separately with $E = 1/2/5$, and the corresponding loss curves and quantitative assessments are shown below.

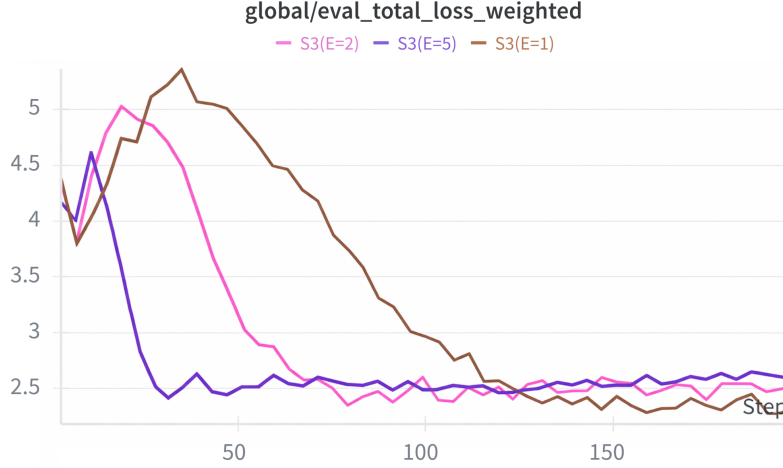


Figure 5: Comparison of convergence trajectories under different local update steps.

Table 2: Performance comparison under different local update steps E .

Setting	FID ($\downarrow$)	IOU ($\downarrow$)
E=1	21.5906 ± 8.1046	0.0186 ± 0.0630
E=2	29.3795 ± 9.9534	0.0148 ± 0.0596
E=5	36.8420 ± 8.5149	0.0229 ± 0.0669

The empirical findings indicate that the local update step number E governs both the stability of federated training and the quality of generated layouts. When $E = 1$, the convergence curve is smoother and achieves the lowest FID score. As local update intensity increases, the models with $E = 2$ and $E = 5$ reach local minima more quickly but subsequently exhibit divergence. The FID scores rise to 29.3795 and 36.8420, respectively, indicating that distributional divergence between synthesized scenes and the real data manifold intensifies. By contrast, IOU remains relatively stable across different update steps, suggesting that the effect of local steps on geometric overlap consistency is weaker than its influence on overall distributional quality.

From the perspective of RQ3, these results suggest that collaborative effectiveness in federated generative layout learning is highly sensitive to local update intensity. Smaller local update steps better preserve inter-client coordination and reduce the risk of client drift under fixed communication budgets. Based on this observation, the subsequent main experiments adopt

$E = 1$ as the default configuration to reduce interference from system parameters in convergence behavior and isolate the effect of data heterogeneity itself.

4.4. Assessing Functional Heterogeneity and Layout Quality across Partition Scenarios

To evaluate how different forms of client heterogeneity alter optimization behavior and generation performance, we compare the convergence outcomes and quality metrics under the five partition settings S0 to S4.

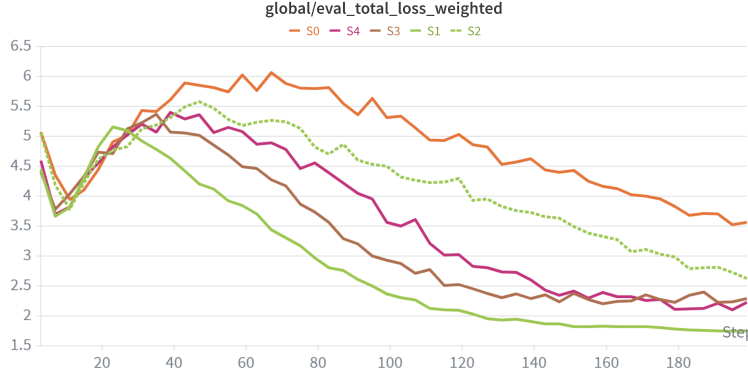


Figure 6: Comparison of convergence trends among different partition settings under the 50-step training budget.

Figure 6 shows that different forms of client heterogeneity do not affect federated generative layout modeling in the same way. Under the same 50-step training budget, S1 (label skew) achieves the lowest weighted residual of 1.74, indicating the most favorable convergence among all groups. By contrast, S2 (feature skew) reaches a substantially higher terminal residual of 2.62, while S3 (quantity skew) and S4 (compound skew) remain in an intermediate range, with weighted residuals of 2.28 and 2.23, respectively. S0 (baseline) records the highest final residual of 3.55, suggesting that a mixed distribution is not necessarily the easiest regime for optimization in the present generative setting.

This tendency is further reflected in the generation-quality evaluation shown in Table 3.

The FID results are broadly consistent with the convergence trends. S1 (label skew) attains the best distributional fidelity, with the lowest FID of

Table 3: FID and IOU results under different partition settings.

Group	FID ($\downarrow$)	IOU ($\downarrow$)
S0	27.1607 ± 12.7905	0.0154 ± 0.0599
S1	12.6994 ± 3.0890	0.0135 ± 0.0517
S2	32.2228 ± 8.8801	0.0150 ± 0.0499
S3	21.5906 ± 8.1046	0.0186 ± 0.0630
S4	16.9751 ± 5.9230	0.0189 ± 0.0625

12.6994, whereas S2 (feature skew) records the worst FID of 32.2228. S3 (quantity skew) and S4 (compound skew) outperform the homogeneous baseline S0, achieving FID values of 21.5906 and 16.9751, respectively, compared with 27.1607 for S0 (baseline). These results suggest that feature-space heterogeneity imposes a stronger obstacle to global distribution fitting than label imbalance or quantity skew in the present task.

At the same time, the IOU values remain relatively close across groups, ranging from 0.0135 to 0.0189. This indicates that the heterogeneous settings differ more substantially in distributional realism than in coarse overlap-related geometric regularity. In other words, client heterogeneity in this task appears to affect the model more strongly at the level of global layout distribution than at the level of simple collision-related statistics.

In relation to RQ1, the results indicate that room-type-oriented label skew is more readily accommodated than style-induced feature skew. In response to RQ2, the same pattern suggests that stylistic heterogeneity is more detrimental to the fidelity of synthesized layout distributions than to basic overlap-related feasibility.

5. Discussion

5.1. Assessing the Impacts of Functional Heterogeneity

RQ1 aims to investigate how room type-induced label shift and style-induced feature shift affect the training behavior and convergence characteristics of federated generative models.

From the perspective of practical deployment, this distinction is meaningful. Room-type label skew is consistent with the situation in which different companies or teams collect data from different service domains, such as bedroom-centric, living-room-centric, or dining-room-centric projects. By

contrast, style-based feature skew better reflects data heterogeneity among organizations that serve similar room types but differ in market positioning, regional preference, or cultural design preference.

The experimental results show that the optimization difficulty of federated layout generation depends not only on whether client data are Non-IID, but also on the form of the underlying distribution shift. In particular, style-induced feature skew causes greater optimization difficulty for the generative model than room-type-oriented label skew. A plausible explanation is that the present generator performs category-wise alignment during reconstruction, which helps preserve reusable object-level regularities even when room-type proportions differ across clients. In contrast, the feature-skew setting alters continuous spatial organization patterns, such as compactness, spacing, and relative arrangement density, thereby making it harder for the global model to reconcile local client objectives through parameter averaging alone. Therefore, data heterogeneity caused by style shift leads to a more challenging optimization landscape for standard FedAvg-based coordination.

5.2. Assessing Layout Distribution Quality under Heterogeneity

RQ2 concerns how room-type variation and style differences affect the quality, plausibility, and diversity distribution of synthesized layouts. The results show that heterogeneity has a more pronounced effect on FID than on IOU, suggesting that its influence is stronger at the level of global distributional realism and overall distribution consistency than at the level of coarse overlap-based feasibility. In other words, layouts generated under feature-skew heterogeneity may still satisfy basic collision constraints, while remaining distant from the target manifold of real design distributions.

This pattern is consistent with the model architecture used in this study. Category-wise linear assignment and reconstruction within semantic groups help mitigate the influence of room-type proportion differences. By contrast, the relational branch explicitly models pairwise translation, rotation, size similarity, and vertical offset, which are more directly associated with continuous layout style. Therefore, room-type heterogeneity does not necessarily degrade generation quality and may even improve fitting under limited training budgets when the generated sub-distribution is more concentrated. Style-induced feature skew, however, is the more critical factor affecting layout realism and distributional consistency, even when basic geometric feasibility remains relatively stable.

5.3. Assessing Collaborative Effectiveness under Practical Constraints

RQ3 asks to what extent federated training can effectively integrate data contributions from diverse clients under practical constraints.

The local-update experiment provides direct evidence on this issue. When the local update step is set to $E = 1$, the model achieves the best FID and the smoothest convergence. Increasing the local update intensity to $E = 2$ and $E = 5$ raises the FID substantially, while IOU changes only slightly. This indicates that larger local steps mainly amplify global distributional drift rather than substantially altering low-level overlap consistency. Therefore, under the current optimization setting, smaller local update steps appear more suitable for preserving collaborative effectiveness.

Additional evidence can also be drawn from the longer-horizon convergence behavior under different partition settings.

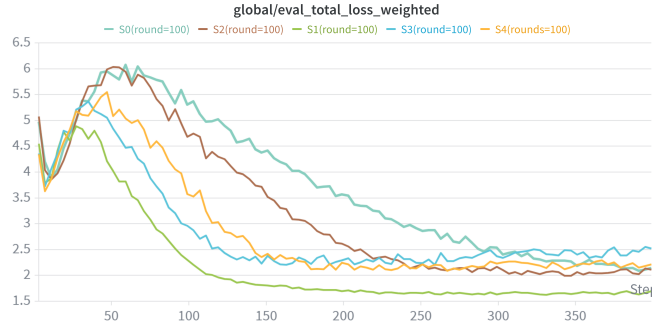


Figure 7: Comparison of convergence trends among different experimental groups under the 100-step training budget.

Figure 7 shows that after increasing the number of training rounds, the trajectory of S0 continues to decline, indicating that the homogeneous baseline still retains room for further fitting under prolonged training. By contrast, S1–S4 gradually enter a plateau phase and remain relatively stable. This divergence in trajectory suggests that simply extending communication rounds is not equally beneficial across different heterogeneity regimes.

A likely interpretation is that when client objectives remain relatively aligned, as in the more homogeneous setting, additional rounds can continue to refine the shared global solution. However, under stronger heterogeneity, the local objectives of different clients may become systematically misaligned, so that further communication yields diminishing returns. In such cases, the

plateau does not necessarily imply full convergence to an optimal shared solution; rather, it may reflect the limited ability of standard FedAvg to reconcile persistent client-specific drift.

Overall, the answer to RQ3 is that federated generative layout modeling remains feasible under moderate heterogeneity and realistic communication constraints, but its collaborative effectiveness depends strongly on the form of the client-data shift and on keeping local training intensity under control. In particular, simple FedAvg coordination can already support beneficial collaboration when clients differ mainly in room-type composition, whereas stronger style-driven divergence is more likely to require heterogeneity-aware aggregation, personalized modules, or explicit alignment strategies for relational spatial features.

6. Conclusion

This paper investigated how Non-IID client-data heterogeneity affects federated generative modeling for automated 3D furniture layout generation. Motivated by the practical need for cross-institutional collaborative model training without sharing commercially sensitive layout data, we formulated the task within a privacy-preserving federated framework and examined two representative sources of heterogeneity: room-type-based label shift and style-based feature shift.

The empirical results provide three main conclusions. First, in response to **RQ1**, different forms of heterogeneity do not affect federated optimization equally. Room-type-induced label skew is comparatively easier to accommodate, whereas style-induced feature skew imposes a stronger challenge to convergence. Second, in response to **RQ2**, feature-space heterogeneity affects the realism and distributional quality of synthesized layouts more strongly than coarse overlap-related feasibility, which is reflected by the larger variation in FID than in IOU. Third, in response to **RQ3**, federated collaboration remains effective under practical constraints when local update intensity is properly controlled, with smaller local update steps yielding more stable convergence and better global fitting.

The study also has several limitations: although the current framework reveals the influence of different heterogeneity patterns, it does not yet include a dedicated mechanism for mitigating plateau behavior under strong client divergence. In particular, when style-induced feature skew dominates

client differences, simply increasing communication rounds or prolonging local training is unlikely to provide a robust solution. Future work should therefore consider heterogeneity-aware aggregation, personalized modules, or explicit alignment strategies for relational spatial features.

Overall, this study demonstrates that federated learning can help break data silos in the interior furnishing domain and support privacy-preserving generative modeling across institutions. At the same time, the results show that the success of such collaboration depends strongly on the type of heterogeneity being integrated, especially because style-based feature skew directly interferes with the continuous relational geometry that governs layout plausibility. These findings provide empirical support and practical guidance for future distributed layout modeling systems in privacy-sensitive design environments.

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